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Does my rating give me more credit?

A panel-data analysis of the impact of credit rating changes on the capital structure of North American firms

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Does my Rating give me more Credit?

A panel-data analysis of the impact of credit rating changes on the capital structure of North American firms

Caroline Hyldahl & Ida Blegvad Mortensen

Abstract

The purpose of this paper is to investigate the effects of real and probable credit rating changes on firms' capital structure decisions. By using a panel fixed effects regression model based on 2,045 North-American firms from 1985-2017, we find that probable and real credit rating changes lead to diverging results. The credit rating-capital structure (CR-CS) hypothesis by Kisgen (2006) holds to some extent, such that probable and real downgrades lead to firms issuing less net debt relative to equity. However, in contrary to the CR-CS theory, our results show that firms after a real upgrade will issue more debt to reap the benefits, and when facing a probable upgrade firms do not adjust debt significantly. We examine if our results are conditional on certain firm-level attributes, and find that investment grade firms and commercial paper issuing firms react stronger to probable and real credit rating changes than their counter-parties. Overall, our findings indicate that credit ratings are a factor, which firms consider when deciding on capital structure. We find the research highly relevant in regards to the current world situation with a possible recession.

Keywords: Credit ratings, capital structure, debt versus equity issuance, probable and real rating changes, broad ratings, notch ratings.

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1 Introduction

Deciding on a capital structure is very central to any firm's corporate division as it is fundamental to firm value. Prudent CFOs have at least one eye permanently fixed on a firm's mix of debt and equity; they must constantly weigh the benefits of debt over equity against the risk of financial distress and bankruptcy. Kisgen (2006) was the first to formally investigate the impact of credit ratings on capital structure decisions, and he developed the credit rating-capital structure (CR-CS) hypothesis which argues that firms close to a credit rating change will issue less debt due to discrete costs and benefits associated with each credit rating level. In reality, managers also seem to take credit ratings into account when making capital structure decisions today. For example, Bloomberg (2019) reported that after the credit rating agency, Standard & Poor's, put The Kraft Heinz Company on notice to cut debt relative to earnings, the company was *"fully committed to deleveraging in a timely manner and to maintaining an investment grade rating going forward"*. The motivation for the interest in the topic can furthermore be traced back to Graham and Harvey's (2001) survey paper which finds that maintaining a credit rating is the second most important objective in a firm's capital structure decisions. They add that credit rating rank higher than many factors suggested by traditional capital structure theories such as profitability and tax shield.

Credit ratings seem even more important today for firms' value and thus capital structure due to the current world situation. In the latest World Economic Outlook on October 11th 2022, IMF warned that *"the worst is yet to come"* for the global economy. Consequently, CFOs are experiencing the most expensive credit since the financial crisis as well as a decreasing availability of credit according to the Deloitte CFO survey 2022 Q3 (Deloitte, 2022). During the financial crisis, tight credit supply was found to have accounted for a substantial part of the fall in GDP (Barnett & Thomas, 2013). This indicates how important credit supply is for the economic activity, and also how important credit supply might be to overcome the potential recession the world economy is heading towards. A way of increasing the individual firm's access to credit can be by possessing a good credit rating. Credit ratings do therefore not only serve to inform investors about firms' creditworthiness but can also serve as a tool to help firms improve their capital structure and navigate in a tight economy.

Credit ratings, as a determinant of capital structure, have not been studied sufficiently in recent years, and there is not a consensus in the literature regarding the relationship between credit ratings and capital structure (Aktan, Çelik, Abdulla, & Alshakhoori, 2019). This study thus contributes to the theoretical and empirical capital structure decisions framework by addressing the research question:

How does credit rating changes affect the capital structure of North-American firms?

We assess that our research can be informing and valuable for different market actors such as CFOs, advisory firms, investment banks, and rating agencies which will be discussed further.

We use a panel fixed effects regression model based on 2.045 North-American firms to expand Kisgen's (2006) research by including the time period from 1985-2017. Furthermore, we explore the different effects from probable and real credit rating changes, respectively, on capital structure inspired by Krichene and Khoufi (2015), and we investigate if our results hold more strongly for certain firm-level attributes as Kemper and Rao (2013). Our research shows that both probable and real credit rating changes have significant effects on capital structure decisions. The CR-CS hypothesis holds to some extent such that probable and real downgrades lead to firms issuing less debt relative to equity. However, the CR-CS hypothesis does not hold when looking at upgrades; after a real upgrade firms will issue significantly more debt, whereas firms facing a probable upgrade will not adjust their capital structure significantly. Furthermore, when examining certain firm-level attributes our findings show that investment grade and commercial paper issuing firms react significantly stronger to rating changes than their counter parties. The effects from rating changes on capital structure for firms with higher external financing needs and larger growth opportunities are not as significant.

The paper is structured as follows. Section 2 outlines the theoretical motivation of the paper concerning credit ratings and capital structure. Section 3 describes the empirical methodology and the research data. Section 4 provides the results of our empirical analysis, and section 5 discusses how our results can be applied to the real world, and evaluates our data-set. Finally we conclude our research in section 6.

2 Theoretical motivation

This section provides the theoretical framework behind credit ratings and capital structure. Firstly, credit ratings and its importance for capital structure are described. Secondly, the traditional theories on capital structure are outlined and lastly, we will explain the credit rating-capital structure (CR-CS) theory, which emphasizes the impact of credit ratings on capital structure decisions.

2.1 What are credit ratings?

Credit ratings are the opinion from a credit rating agency of the credit risk of an entity. They are based on the forward looking probability that firms will meet its financial commitments continuously and they also rate the quality of firms' individual debt issuance (S&P, 2022a). There are multiple well-recognised rating firms, and the three largest global agencies are Moody's, Fitch, and Standard & Poor's (S&P). We will use the long-term credit rating from S&P, which consists of 22 ratings from AAA to D that the agency assigns to corporate and sovereign debt. We have chosen to use the credit ratings from S&P, since this is standard in the literature, and S&P is considered the largest of the Big Three credit rating agencies. Table 1 below shows S&P's long term issuer credit ratings and the classification of either investment or speculative grade, and the numerical coding of broad and notch ratings, which will be used later in the empirical analysis.

The term "investment grade" consists of ratings from AAA to BBB-, and is used to describe relatively high levels of creditworthiness and credit quality such that the firm is able to meet its debt obligations with no concerns. In contrast, debt issuers receiving a rating lower than BBB- are considered to be "speculative grade" or "high yield". This refers to debt issuers who currently have the ability to repay but faces significant uncertainties such as financial distress, which could affect their credit risk (S&P, 2022b). Broad rating classification is a way of dividing the credit rating scale into more groups. Investment grade firms are divided into three segments, and speculative grade into five groups such that group 1 is a pool of the best rated firms. Most broad ratings have notches (+/non/-) that further divide the broad rating groups into subcategories to refer to the relative position within each category (Aktan

et al., 2019) which makes notch rating the most segmented division of the scale.

Table 1: S&P long term issuer credit ratings

Rating	Classification (IGSG)	Broad rating (BR)	Notch rating (NH)
AAA	Investment grade	1	1
AA+		1	2
AA		1	3
AA-		1	4
A+		2	5
A		2	6
A-		2	7
BBB+		3	8
BBB		3	9
BBB-	Speculative grade	3	10
BB+		4	11
BB		4	12
BB-		4	13
B+		5	14
B		5	15
B-		5	16
CCC+		6	17
CCC		6	18
CCC-		6	19
CC		7	20
C		7	21
D		8	22

Note: The table shows the relation between the S&P long term issuer credit ratings and the classification of either investment or speculative grade, and the numerical coding of broad and notch ratings.

2.2 The importance of credit ratings for capital structure

There are various reasons why credit ratings could have a direct impact on a firm's capital structure, and so far there is not a consensus in the literature concerning the precise relationship, but there are several suggestions.

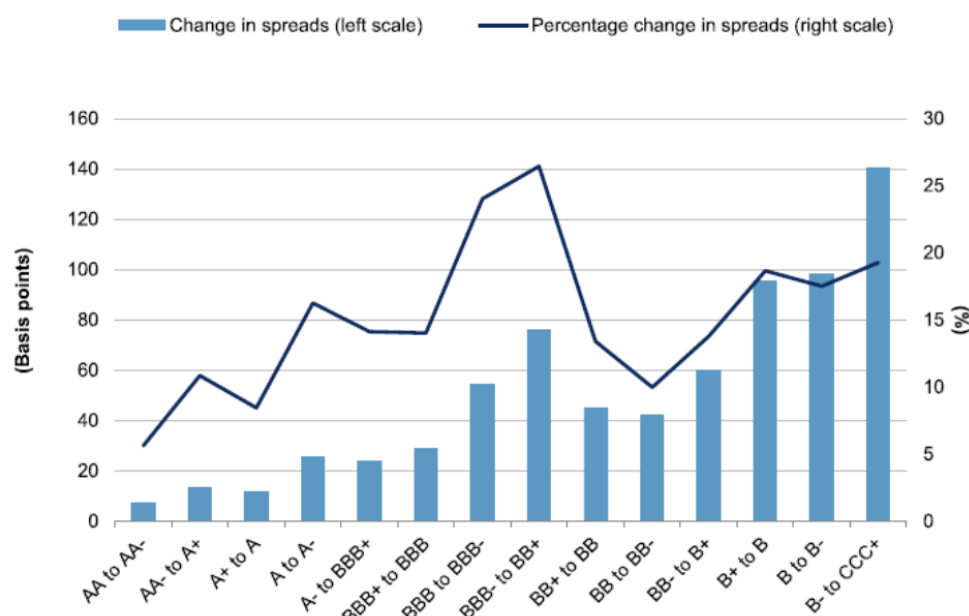
What Kisgen (2006) suggests to be a driving factor of the relationship is first of all that there are regulations on bond investments that are tied directly to credit ratings. Capital requirements related to investments, by for example insurance firms, are dictated by the credit rating (Kemper & Rao, 2013). Banks are also restricted from owning speculative-graded bonds, and pension fund guidelines are also often restricted to only investment-grade bonds. Regulations due to credit ratings can also have impact on the liquidity on the bonds; higher yield bonds in less liquid markets might be avoided by investors, and these speculative

grade bonds can thus have difficulties financing their debt in economic downturns, which increases the costs on the firms further. In general, notch rating changes do not give rise to significant regulations, so in theory we would not see large effects on the capital structure due to regulations when looking at a notch credit rating change compared to a broad rating change or especially a change between investment and speculative grade.

Second of all, credit ratings serve as a smoothing function on the capital markets by conveying information to the market about the quality of a firm that is not public knowledge. Information that serves as a supplement to public available information can be crucial in any evaluation process for investors, hence credit ratings might be able to reduce the information asymmetry between the firm and investors. Bosch and Steffen (2011) find that the information produced by the rating agencies have an important link to how ratings can improve access to capital. Furthermore, Pan, Lin, Lee, and Ho (2015) conclude that information asymmetry is an important factor for firms' financial decision-making and less information asymmetry leads to less net debt issuance relative to equity as the cost of issuing equity will then be lower than the cost of issuing debt. Information should have impact on capital structure for any credit rating up- or downgrade, also a notch rating change (Kisgen, 2006) unlike bond regulations. This is due to the fact that if ratings convey information then firms with the same credit rating will be pooled together, and it will signal that they have the exact same firm quality and default probability. However, in reality there might be marginal differences so firms will have an incentive to adjust their capital structure in order to be pooled with the firms in a higher notch rating class or avoid being pooled with firms in a lower notch rating category.

The third argument by Kisgen (2006) is that there is a direct cost/benefit to the firm related to a specific rating. First of all, the bond yield will rise for every notch rating downgrade, which makes it more costly for firms to issue debt. Figure 1 below shows how spreads increase as ratings drop, and the largest gaps are between investment and speculative grade (from BBB- to BB+), and between the lowest speculative grade ratings.

Figure 1: Change in spread for every notch downgrade



Notes: Date as of February 13, 2019. Source: S&P Global Fixed income research (S&P, 2019). The figure does not show the change in spread for the highest and lowest credit ratings.

These discrete costs/benefits will thus have an effect on capital structure for any notch rating change, and in theory we would expect to see the largest effect from a change between investment and speculative grade, since the spread - and thus the cost - will increase by more than 25%. We would also expect broad rating changes to have a larger impact on capital structure than a notch rating change. Furthermore, a firm's credit rating can impact and limit operations, contracts and the types of investors who will buy the bonds. For example, merger deals are sometimes dependent on the firms involved maintaining their credit ratings (Kisgen, 2007), and firms entering into long-term supply contracts may require specific credit ratings from their counter-party to avoid the risk of default. Downgrades could also have a negative impact on employees and customers, that can pass on negative effects to the daily operations. Firms facing a real downgrade can further face costly events known as "rating triggers" which for example can be a required repurchase of bonds. Access to other financial markets are also conditional on the long-term credit ratings such as the commercial paper (CP) market (Anderson & Gascon, 2009). Companies issue CP as a low-cost alternative to bank loans, and these issues are not typically backed by any specific collateral thus the

lenders must rely on the firms' financial ability to repay this obligation. A firm can have a CP rating with A1 as the highest rating (using S&P's notation), and according to S&P a long-term rating of A or higher is required to gain a CP rating of A1 (Kisgen, 2007). For these CP-issuing firms, there might occur greater costs (benefits) of downgrades (upgrades) which implies that ratings are more important for their capital structure decisions.

It is therefore evident that there is a relationship between credit ratings and capital structure, but at the same time many other factors can impact the decisions on optimal capital structure for a firm. We will now outline the traditional theories concerning capital structure, which does not take credit ratings explicitly into account.

2.3 Traditional theories on capital structure

Modigliani and Miller's (1958) theory was the first formal treatment of capital structure, and they demonstrated that capital structure (the debt-equity mix) is irrelevant for the firm given certain capital market assumptions. However, they did not take the presence of a tax shield into account so they corrected their formulation by including a tax shield benefit (Modigliani & Miller, 1963), and argued that a levered firm's value would be equal to the sum of an unlevered firm's value plus the tax shield benefit. It indicated that heavy leveraged firms would be more valuable than their counterparts, which was a problem caused by not taking financial bankruptcy costs into account. Two theories named trade-off theory and pecking order theory later emerged and tried to deal with this problem when explaining capital structure decisions. The trade-off theory proposes that a value-maximizing firm will balance the benefits of debt (from the interest tax shield) with the costs of debt (bankruptcy and financial distress) to determine an optimal leverage for the firm (Fama & French, 2002). On the other hand, Myers' (1984) pecking order theory argues that firms will in general fund projects first with internal financing, and if external financing is needed, they will prefer debt over equity due to asymmetric information costs associated with the latter.

A traditional view on the supply side of debt have thus been that it was perfectly elastic, so capital structure decisions depended solely on the demand for debt as the above theories describe (Judge & Korzhenskaya, 2022). It has later become evident that because of market

imperfections as information asymmetry and regulations, there exist credit constraints that might have impact on a firm's capital structure decisions. The credit rating-capital structure (CR-CS) model developed by Kisgen (2006) was the first to focus on the supply side of debt as a determinant of capital structure.

2.4 The credit rating-capital structure model

According to the CR-CS hypothesis, credit ratings is an important factor to consider in corporate finance decisions due to the discrete cost or benefit associated with different rating levels. This means that firms facing a probable rating change will issue less debt in order to achieve an upgrade or to avoid a downgrade. Kisgen used a sample of North-American firms in the time period from 1986-2001, and when empirically testing this hypothesis, the study could conclude that firms near credit rating changes will issue app. 1% less net debt relative to equity compared to firms not near a credit rating change.

After Kisgen proved a relationship between credit rating and capital structure, other researchers replicated and extended the analysis which have yielded ambiguous results. Krichene and Khoufi (2015) investigates the relationship on a sample of American firms listed on the S&P500 index from 2007-2010, and they find similar results to Kisgen. Namely, that firms close to an upgrade to investment grade or close to a downgrade to speculative grade will issue less debt relative to equity to either avoid a downgrade or increase the chances of an upgrade. On the other hand, Kemper and Rao (2013) concluded that the CR-CS model does not apply to all rating levels and argued that Kisgen's results are driven by the subsample of firms with low credit ratings. Furthermore, they could not reject their hypothesis that the CR-CS model is conditional on certain firm attributes.

In follow-up studies, Kisgen (2007, 2009) examines how corporate financial decision-making is affected by *real* credit rating changes. The findings indicate that downgraded firms will issue significantly less net debt relative to equity. The effect of a downgrade is larger at downgrades to speculative grade, and if commercial paper access is affected. However, rating upgrades do not seem to affect capital structure decisions, which suggests that firm target minimum credit rating levels. Krichene and Khoufi (2015) also examines the effect of real credit rating changes

on capital structure decisions, and they find like Kisgen that firms receiving a downgrade to speculative grade will reduce their net debt issuance in order to reach investment grade again. The paper further concludes that once a firm has been upgraded to investment grade, it will issue more debt without the fear of being downgraded. Krichene and Khoufi thus find that capital structure reactions to probable and real credit rating changes are not similar. Aktan et al. (2019) uses a sample of all non-financial listed firms on the Saudi Arabia stock exchange during the period 2009 to 2016. This study examines both credit rating changes to investment grade or speculative grade, broad rating changes and notch rating changes. Their results are divergent, but in general their findings do not support the CR-CS hypothesis. Finally, Huang and Shen (2015) investigate the relationship between capital structure adjustment and credit rating changes based on a global cross-country sample. Like Kisgen (2007, 2009) their results demonstrate asymmetric effects such that firms adjust their capital structure when ratings are downgraded, but do not significantly adjust their leverage ratios when ratings are upgraded. This is again aligned with the theory about firms targeting minimum credit ratings and not following CR-CS behavior. We will now look at our empirical approach for the research.

3 The empirical methodology

This section seeks to give a brief introduction to the empirical methodology used in our analysis and an overview of the employed data.

3.1 Panel data and fixed effects regression model

Our research is conducted with panel data, which combines time-series data and cross-sectional data. In our analysis we have missing years for many of the firms in our sample, and we thus have an unbalanced panel. We will outline the method we decided was best for our research, the fixed effects estimation, and we will argue why it fits our data better than other methods (Pooled OLS, First Differencing and Random Effects model). The outlining of the methods follows Wooldridge (2015).

The most basic method for estimating panel data is the the Pooled OLS, which is used by Kisgen(2006) and in many other research papers. However, using this method can result in

biased and inconsistent estimates if the unobserved firm-specific fixed effects, a_i are correlated with the explanatory variables, x_{it} from the original unobserved model outlined below. The zero correlation assumed in the Pooled OLS estimation is not realistic in our panel data set-up, and we have thus chosen not to use this method. This is also why we do not find the Random Effects estimation appropriate to use, since the method has the assumption as well that a_i is independent of all explanatory variables in all time periods. We are thus seeking a method to eliminate the fixed effect, a_i . The two most used methods to do this are Fixed Effects (FE) or First Differencing (FD).

With the FD method, one first-differences panel data over time in order to eliminate the time-constant unobserved effect, a_i . However, since our estimation is based on an unbalanced panel the FE has an advantage over FD. Namely, FE maximizes the number of observations used in the estimation, whereas FD only uses a time period t if time period t and $t - 1$ both include a full set of observations. The FD method would thus eliminate a relatively large amount of observations in our unbalanced panel data-set, and we have thus chosen the FE estimation in our analysis.

FE estimation is useful under the assumption that the unobserved time-invariant firm-specific effects are correlated with the regression variables. The original unobserved model is outlined as:

$$y_{it} = \beta_1 x_{it1} + \beta_2 x_{it2} + \dots \beta_k x_{itk} + a_i + u_{it}$$

where y_{it} is the dependent variable, $x_{it1}, x_{it2}, \dots, x_{itk}$ for $t = 1, 2, \dots, T$ are the explanatory variables, a_i are the unobserved effects and u_{it} are the idiosyncratic errors. For each i we then average this equation over time, which yields:

$$\bar{y}_i = \beta_1 \bar{x}_{i1} + \beta_2 \bar{x}_{i2} + \dots + \beta_k \bar{x}_{ik} + a_i + \bar{u}_i$$

Because a_i is fixed over time, it appears in both equations. We then subtract the two equations for each t , and end up with the time-demeaned model:

$$y_{it} - \bar{y}_i = \beta_1 (x_{it1} - \bar{x}_{i1}) + \beta_2 (x_{it2} - \bar{x}_{i2}) + \dots + \beta_k (x_{itk} - \bar{x}_{ik}) + u_{it} - \bar{u}_i$$

$$\tilde{y}_{it} = \beta_1 \tilde{x}_{it1} + \beta_2 \tilde{x}_{it2} + \dots + \beta_k \tilde{x}_{itk} + \tilde{u}_{it}$$

This transformation has thus resulted in the elimination of the unobserved effect, a_i , that probably had been correlated to the explanatory variables and thus led to a biased model. However, we still have one key assumption that is crucial for the model to be unbiased, which is that the idiosyncratic errors should be uncorrelated with the explanatory variables so: $\text{cov}(x_{itj}, u_{it}) = 0$ for all i , t , and j , which indicates that the explanatory variables are strictly exogenous after eliminating the unobserved effects a_i . The assumption is in general difficult to fulfill, but the same assumption would apply to other panel data estimation models as well. Furthermore, robust standard errors are used in our estimation to account for any heteroskedasticity in the standard errors and we perform a Hausmann-test to ensure FE is the preferred model compared to RE.

3.2 Data and summary statistics

3.2.1 Sample and sampling method

The sample is constructed by first extracting data for all North-American firms in the Compustat database. The data-set is merged on another data-set consisting of all North-American firms with a credit rating in Compustat at the beginning of a year. The credit rating used is S&P's Long-Term Domestic Issuer Credit Rating, which is described in section 2.1. The sample period is 1985 to 2017, since this is the period for which the credit rating is available in Compustat. In the construction of the sample, we exclude firm-years where the firm has missing data of the variables, which are required in order to test our hypotheses in the regression analysis¹. This is done to ensure high data quality. We also ensure that there are no doublets of firm-years in the data sample, since this is required for a panel data set-up. Previous papers such as Fama and French (2002) and Judge and Korzhnitskaya (2022) often exclude utilities and financial companies, since they argue that these companies have a significant different appetite towards debt. However, Kisgen (2006) shows robustness to their exclusion from the sample, and we thus choose to include these type of firms. The final sample in our baseline regression model consists of 2.045 firms with 24.111 firm years.

¹These are all the variables described in table 2 and in appendix except the variable "Changes in current debt for firm i from time $t-1$ to t " used for the variable ΔD_{it} .

3.2.2 Definition of variables

Table 2 is an overview of the variables used in our research. The variables for probable and real credit rating changes are based on Kisgen (2006) and Aktan et al. (2019). The independent variables used when extending our analysis in section 4.2 are based on Kemper and Rao (2013), Karakaya, Kurtaran, Kurtaran, et al. (2017), and Chung and Pruitt (1994). Our choice of control variables follows Kisgen (2006) and Judge and Korzhnitskaya (2022), since they have found leverage, profitability, and size to influence capital structure choice and to mitigate problems of omitted variable bias.

Table 2: Variables

Variable	Explanation
<i>Dependent variable</i>	
$NetDIss_{it} = (\Delta D_{i,t} - \Delta E_{i,t})/A_{i,t}$	The firms' issuance of debt relative to equity. Net debt minus net equity raised during the year, divided by total assets.
<i>Independent variables for a probable rating change (dummy variables)</i>	
$CR_{POM} = CR_{Plus} + CR_{Minus}$	Firms with a minus or plus credit rating at the beginning of period t .
CR_{Plus}	Firms with a plus credit rating at the beginning of period t .
CR_{Minus}	Firms with a minus credit rating at the beginning of period t .
<i>Independent variables for a real rating change (dummy variables)</i>	
$IGSG_{up}$	Firms with an upgrade from speculative to investment grade from period $t - 1$ to t .
$IGSG_{down}$	Firms with a downgrade from investment to speculative grade from period $t - 1$ to t .
BR_{up}	Firms with a broad credit rating upgrade from period $t - 1$ to t .
BR_{down}	Firms with a broad credit rating downgrade from period $t - 1$ to t .
NH_{up}	Firms with a notch credit rating upgrade from period $t - 1$ to t .
NH_{down}	Firms with a notch credit rating downgrade from period $t - 1$ to t .
<i>Independent variables used in section 4.2</i>	
EFN = Real growth rate of assets – sustainable growth rate	A variable (proxy), which captures the firms' external financing needs (EFN). We remove all observations with $EFN > 1$, since it is unlikely that a firm's financing needs are greater than its assets.
Tobin's $Q \approx \frac{MVE + PS + DEBT}{TA}$	An approximation of Tobin's Q to assess firm's growth opportunities. MVE is the market value, PS is the liquidating value of a firm's outstanding preferred stock, $DEBT$ is the value of the firm's short-term liabilities net of its short-term assets plus the book value of the firm's long term debt, and TA is book value of total assets.
<i>Set of control variables, K_{it}</i>	
$Lev = D_{i,t-1}/(D_{i,t-1} + E_{i,t-1})$	The firm's leverage.
$Prof = EBITDA_{i,t-1}/A_{i,t-1}$	The firm's profitability.
$Size = \ln(Sales_{i,t-1})$	The firm's size based on their sales
<i>Note: The variables used to construct the dependent and independent variables are further explained in the appendix.</i>	

3.2.3 The difference between a probable and real rating change

The difference between a probable and real credit rating change is crucial for understanding the difference in the two baseline models in section 4.1.1 and 4.1.2. We categorize firms as facing a probable rating change if they are near a rating change. This is the case if their credit rating is designated with a "+" or "-", and firms are on the other hand not near a rating change if they do not have a plus or minus notch, for example BB compared to BB+ or BB-. We will discuss the shortcomings of this proximity measure in section 5.1. On the other hand, the independent variables for a real rating change are dummy variables for firms that have already experienced an up- or downgrade in either of the three categories: Notch (NH), broad (BR) or a change to investment or speculative grade (IGSG). We find it interesting to analyze if firms adjust their capital structure differently when facing probable versus real credit rating changes.

3.2.4 Descriptive analysis

Summary statistics for the sample are shown in table 3 and figure 2. Table 3 shows statistics for the net debt issuance by credit rating within the sample, and it also depicts the number of firm years by rating. The table indicates that the sample is relatively well-distributed, since 10 out of the 17 rating categories have above 1,000 firm years. Furthermore, it is evident how the middle ratings have the most firm years, and the tails have the least firm years. This indicates that the sample is characterized by a moderately normal distribution which makes the empirical results in the paper not likely to be driven by any specific credit rating.

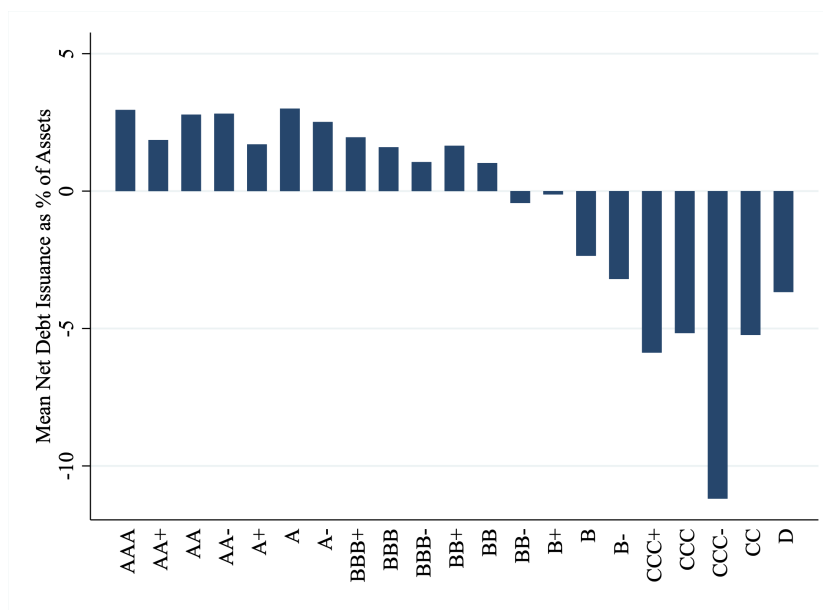
Figure 2 graphically depicts the average net debt issuance relative to equity by rating. The figure shows an indication of a relationship between credit ratings and net debt issuance, namely that firms with higher credit ratings issue more debt and less equity than firms with a poorer credit rating. This indicates that a firm's credit rating situation seems to affect their capital structure decision. Furthermore, table 3 shows how the standard deviation within each rating for the net debt issuance are in general high - at least for the lower credit ratings. This indicates that although there exists a relationship between net debt issuance and ratings, there are significant differences within each credit rating regarding their net debt issuance.

Table 3: Sample summary statistics - Ratings and net debt issuance

	AAA	AA+	AA	AA-	A+	A
Number of firm years	539	214	849	956	1.853	2.749
Net debt issuance						
Mean (%)	2.95	1.86	2.78	2.82	1.70	3.00
Std. dev. (%)	6.51	9.23	5.71	7.53	8.81	8.23
	A-	BBB+	BBB	BBB-	BB+	BB
Number of firm years	2.226	2.432	2.903	1.948	1.117	1.368
Net debt issuance						
Mean (%)	2.52	1.96	1.60	1.06	1.65	1.02
Std. dev. (%)	8.26	7.52	13.77	11.56	11.34	10.97
	BB-	B+	B	B-	CCC+*	
Number of firm years	1.399	1.703	884	445	526	
Net debt issuance						
Mean (%)	-0.44	-0.12	-2.35	-3.20	-6.23	
Std. dev. (%)	11.29	23.35	22.45	33.64	43.43	

Notes: Mean and standard deviation of net debt issuance by credit rating within the sample, and the number of firm years (out of the total sample of 24.111 firm years) that had the indicated rating at the beginning of the firm year. *The last category CCC+ consists of all firms with a credit rating of CCC+ and below.

Figure 2: Capital market activity by rating, 1986-2017



Notes: Average net debt issuance minus net equity issuance by credit rating.

4 Empirical analysis and research findings

In this section we will first of all test if the CR-CS hypothesis still holds in general by replicating Kishin's (2006) regression model using our updated sample of firms covering the period 1986-2017. After looking at the impact of *probable* credit rating changes on capital structure decisions, we will continue by examining the effect of *real* credit rating changes on firms' net debt issuance inspired by Aktan et al. (2019). At last, we will go more into depth with the results of both probable and real ratings change by exploring the heterogeneous effects of credit rating changes across different type of firms as Kemper and Rao (2013).

4.1 Impact of credit rating changes on capital structure decisions

4.1.1 Effect of probable credit rating changes

This section tests whether a firm's net debt issuance for a particular year is affected by a firm being close to an up- or downgrade in their credit rating, which we categorize as having a probable rating change. The CR-CS theory implies that firms with a probable rating change will issue less debt relative to equity than firms not facing at probable change.

The following two regressions test the hypothesis:

$$NetDIss_{it} = \alpha_1 + \beta_0 CR_{POM} + \phi K_{it} + u_{it} \quad (1)$$

$$NetDIss_{it} = \alpha_2 + \beta_1 CR_{Plus} + \beta_2 CR_{Minus} + \phi K_{it} + u_{it} \quad (2)$$

The CR-CS theory predicts that $\beta_i < 0, i = 0, 1, 2$, and the null hypothesis is thus $\beta_i \geq 0$. Table 4 below shows the results of the regressions. The exclusion of the control variables in model 2 and 4 gives an insignificant R-squared, and this shows that the control variables are significant variables, which need to be considered when examining the effect of credit ratings on net debt issuance. We will thus include the control variables in the following regressions as well, but not include the statistics in the tables. Furthermore, we have used a Hausmann-test to ensure that FE is the preferred model to use in our case compared to RE².

²The p-value is 0.000 thus rejecting the null hypothesis that the preferred model is random effects.

Table 4: Effect of probable change in credit rating on the net debt issuance vs. equity

Variables/statistics	Model 1	Model 2	Model 3	Model 4
Intercept	-0.1024***	0.0157***	-0.0570***	-0.0025***
CR_{POM}	-0.0052* (0.0041)	-0.0044 (0.0037)		
CR_{Plus}			-0.0045 (0.0054)	-0.0052 (0.0063)
CR_{Minus}			-0.0049** (0.0029)	-0.0036* (0.0026)
LEV	0.0204		0.0207	
PROF	-0.2819		-0.1916	
SIZE	0.0184***		0.0130***	
F-statistics	6.9000	1.4400	6.5900	1.1500
Prob(F-statistics)	0.0000	0.2297	0.0000	0.3180
R-squared	0.0611	0.0001	0.0556	0.0001

Note: Standard errors are given within the parenthesis. ***, **, and * denote significance at the 1-, 5- and 10-% levels, respectively.

The value for the coefficient β_0 in model 1 indicates that firms with a plus or minus rating annually issue 0.52% less net debt as a percentage of total assets relative to equity than firms not near a change in the credit rating³. This result is qualitatively similar to Kisgen's (2006), and the CR-CS hypothesis holds in this model. However, model 3 analyzes whether there is a larger effect from a probable credit rating downgrade or upgrade on net debt issuance. This shows that the significance of CR_{POM} seems to be large driven by the minus segment i.e. the firms close to a rating downgrade. These firms issue app. 0.49% less debt relative to equity, whereas a probable rating upgrade does not seem to affect capital structure decisions significantly, hence the CR-CS hypothesis does not hold for firms near an upgrade. This suggests that firms target minimum credit ratings over time, which means that they decide on capital structure to avoid being downgraded to a level below their minimum credit rating target and at the same time they are indifferent to being upgraded once they have received this minimum target rating (Kisgen, 2007).

We now want to test the effect on capital structure, when looking at real credit rating changes instead of probable credit ratings changes, since Krichene and Khoufi (2015) find that probable and real credit rating changes do not have the same impact on capital structure.

³This is a one-sided statistical t-test, since the null hypothesis is accepted for any positive values of the coefficient.

4.1.2 Effect of real credit rating changes

We use three models to examine the impact of real credit rating changes on capital structure based on the methodology adopted by Aktan et. al. (2019). First, the effect of a real rating change to investment or speculative grade (IGSG) on the capital structure of the current year is tested. Second, the impact of a real broad rating change (BR) is studied, and thirdly the effect of a real notch rating change (NH) is examined. The regression models to test if the net debt issuance are affected by these types of real rating changes are set up the same way as equation (2) in the previous section such that:

$$NetDIss_{it} = \alpha_3 + \beta_3 IGSG_{up} + \beta_4 IGSG_{down} + \phi K_{it} + u_{it} \quad (3)$$

$$NetDIss_{it} = \alpha_4 + \beta_5 BR_{up} + \beta_6 BR_{down} + \phi K_{it} + u_{it} \quad (4)$$

$$NetDIss_{it} = \alpha_5 + \beta_7 NH_{up} + \beta_8 NH_{down} + \phi K_{it} + u_{it} \quad (5)$$

Again, CR-CS predicts that $\beta_i < 0, i = 3, 4, \dots, 8$, and the null hypothesis is thus $\beta_i \geq 0$. The table below shows our results from the regressions.

Table 5: Effect of real credit rating changes on the net debt issuance vs. equity

Variables/statistics	Model 5	Model 6	Model 7
Intercept	-0.1024***	-0.1234**	-0.1216**
$IGSG_{up}$	0.0285*** (0.0108)		
$IGSG_{down}$	-0.0293*** (0.0104)		
BR_{up}		0.0148** (0.0065)	
BR_{down}		-0.0250*** (0.0091)	
NH_{up}			0.0021 (0.0060)
NH_{down}			-0.0231*** (0.0073)
F-statistics	9.3300	9.3900	14.6300
Prob(F-statistics)	0.0000	0.0000	0.0000
R-squared	0.0597	0.0602	0.0606

Note: Standard errors are given within the parenthesis. ***, **, and * denote significance at the 1-, 5- and 10-% levels, respectively.

From table 5 it is evident that the effects from a real credit rating change are not similar to the above effects of a probable credit rating change on capital structure decisions aligned with Krichene and Khoufi (2015). Overall, firms issue less net debt following a downgrade, and

they issue more debt following an upgrade. The size of the coefficient further indicate that a downgrade has a larger effect on the net debt issuance compared to an upgrade. This also shows that the CR-CS hypothesis does not hold when examining real credit rating changes.

The impact on net debt issuance of being upgraded to investment grade ($IGSG_{up}$) or downgraded to speculative grade ($IGSG_{down}$) are both very significant at a 1% level. These results show that firms downgraded to speculative grade issue 2.93% less net debt compared to firms who have not seen a change in their credit rating. This can be explained by the fact that a speculative credit rating limits the firm's ability to issue debt. On the supply side, this means that regulators use credit ratings as a threshold to determine whether an institutional investor may buy the bonds of a certain company. Furthermore, regulations do not allow for example banks, pension funds and insurance companies to take position in speculative grade bonds (Kemper & Rao, 2013) as described in section 2.2. This loss of investors on the capital market probably contributes to the lower net debt issuance. On the demand side, the downgrade in credit rating will increase costs of debt, which makes firms issue less debt relative to equity. Furthermore, one could argue that firms will probably strive to achieve an investment grade again and thus limit their net debt issuance to increase these chances. On the other hand, the coefficient of $IGSG_{up}$ (β_3) is positive, hence firms that have been upgraded from speculative to investment grade issue 2.85% more net debt. The explanation for this is the same as previously, just the other way around: Fewer regulatory limitations give greater access to the debt capital markets, and the costs of issuing debt are lowered. Since the professional opinion on their risk of default is reduced, the firm will not be concerned about not meeting their debt obligations, which results in greater net debt issuance.

The asymmetric effects from a downgrade and an upgrade can also be seen in model 6 and 7 when examining the impact of broad and notch credit rating changes on firms' net debt issuance. However, the effects are smaller and less significant suggesting that firms are more concerned with credit rating changes to investment or speculative grade than smaller credit rating changes (i.e. broad and notch rating changes), which was expected from theory. Our results furthermore indicate that not only regulations can explain the link between credit rating and capital structure, since a real downgrade in notch rating (NH_{down}) has a significant effect on debt issuance, and regulations do not in general apply to notch rating changes cf.

section 2.2. This means that information asymmetry and discrete costs/benefits also play an important role for financial decision makers when deciding on capital structure.

Our baseline results thus show that the CR-CS hypothesis holds to some extent such that probable and real downgrades lead to firms issuing less net debt relative to equity. However, after a real upgrade the firms will issue more debt, and when facing a probable upgrade firms will not adjust debt significantly which is not aligned with the CR-CS theory. We will now extend our analysis and focus on the heterogeneous effects of credit rating changes on capital structure across different type of firms.

4.2 Heterogeneous effects of credit rating changes

Several studies have hypothesized that some firms react stronger to credit rating changes than others. We will expand the research by Kemper and Rao (2013) and explore four hypotheses concerning which firm attributes make corporations more or less concerned about both probable and real credit rating changes when deciding on capital structure.

4.2.1 Hypothesis 1: Investment versus speculative grade ratings

Our first hypothesis is that the effect from credit rating changes on firms' capital structure decisions differ across bond ratings (H1). More specifically, we split the sample into two sub-samples; one sample with all firms with an investment grade (i.e. firms with a credit rating above BB+), and a second sample consisting of all firms with a speculative grade (i.e. firms with a credit rating below BBB-) to explore if the effects of probable and real credit rating changes apply more strongly to either investment or speculative grade firms. Table 6 shows our findings regarding the impact of probable credit rating changes, whereas table 7 shows the results of real credit rating changes.

The findings in table 6 show the same results for investment grade firms as we saw in the baseline model containing all firms. The CR-CS hypothesis do not hold at all for speculative grade firms hence investment grade firms might be driving our baseline results for probable credit rating changes. This indication is also partly confirmed by the high R^2 for the models that only include investment grade firms compared to the models including speculative grade

firms. We see that firms with an investment grade regulate their capital structure, when being close to an up- or downgrade, whereas speculative grade firms are indifferent. This can be seen from the fact that CR_{POM} for investment grade firms is significantly negative, which is consistent with the CR-CS theory, whereas the same coefficient for speculative grade firms is insignificant. Furthermore, when looking at the effect of either an upgrade or a downgrade, investment grade firms are especially concerned when they have a probable rating downgrade and will thus issue less debt to avoid it, whereas these type of firms do not react significantly to a probable rating upgrade. These results are the same as in the baseline model thus the CR-CS hypothesis does only partly hold in this case as well.

Table 6: Effect of probable credit rating changes on the net debt issuance (investment vs. speculative grade companies)

Variables/statistics	IG	IG	SG	SG
Intercept	-0.1203***	-0.1195***	-0.1511***	-0.1510***
CR_{POM}	-0.0022* (0.0040)		0.0031 (0.0056)	
CR_{Plus}		0.0022 (0.0023)		0.0074* (0.0056)
CR_{Minus}		-0.0060*** (0.0026)		-0.0033 (0.0080)
F-statistics	114.65	95.56	3.5900	4.0200
Prob(F-statistics)	0.0000	0.0000	0.0065	0.0013
R-squared	0.8032	0.8033	0.0023	0.0027

Note: Standard errors are given within the parenthesis. ***, **, and * denote significance at the 1-, 5- and 10-% levels, respectively.

The results in table 7 with real changes also show that it is the investment grade firms, which drive the baseline results in section 4.1.2 on real credit rating changes. As in the baseline scenario, investment grade firms increase their net debt following an upgrade and reduce their net debt following a downgrade. Furthermore, our findings show that real changes in credit ratings, for both broad and notch rating, do not have a significant effect on the net debt issuance for speculative grade firms. One can thus conclude that these type of companies do not take real credit rating changes into account when deciding on their capital structure.

Table 7: Effect of real credit rating changes on the net debt issuance (investment vs. speculative grade companies)

Variables/statistics	IG	IG	SG	SG
Intercept	-0.1355***	-0.1336***	-0.1595***	-0.1510***
BR_{up}	0.0088** (0.0043)		0.0036 (0.0095)	
BR_{down}	-0.0223*** (0.0037)		-0.0078 (0.0117)	
NH_{up}		0.0100*** (0.0027)		0.0052 (0.0063)
NH_{down}		-0.0237*** (0.0024)		-0.0082 (0.0087)
F-statistics	97.19	108.38	4.9400	5.5000
Prob(F-statistics)	0.0000	0.0000	0.0002	0.0001
R-squared	0.8182	0.8187	0.0044	0.0046

Note: Standard errors are given within the parenthesis. ***, **, and * denote significance at the 1-, 5- and 10-% levels, respectively.

The significant difference in the firms' reaction to a real credit rating change can be explained by the difference in the firms' access to the debt capital markets. As mentioned earlier, there are strict regulations for institutional investors to invest in high yield bonds. This means that speculative grade firms will be less concerned with their credit rating, since they are already restricted on the debt market and have fewer opportunities for debt financing than investment grade companies. On the other hand, firms that currently have an investment grade rating are probably more concerned with keeping this position and not becoming a "fallen angel"⁴. This is why these type of firms react so strongly to a downgrade; the coefficients show that they reduce their net debt issuing annually by at least 2% more than firms that are not downgraded. We can thus confirm H1: The effect from credit rating changes on firms' capital structure decisions differ across bond ratings, since investment grade firms react more strongly to both probable and real credit rating changes than speculative grade firms.

4.2.2 Hypothesis 2: External financing needs

Our second hypothesis is that firms with larger external financing needs, thus larger capital market activity, will be more likely to have credit ratings in mind when deciding on capital structure (H2). These firms will possibly be more concerned with the discrete costs associated with credit rating changes than firms who rarely visits the debt capital markets and mostly rely on internal debt financing. This hypothesis can be seen as a potential extension of the

⁴Bonds that were originally investment grade but later get downgraded to high yield status.

pecking order theory, which does not consider the impact of credit ratings on firms' capital structure decisions in relation to external versus internal financing.

The models testing if the net debt issuance is affected more by probable credit rating changes when a firm has larger external financing needs are extensions of equation (1) and (2) from section 4.1.1. We extend the models by including the proxy variable for external financial needs (EFN) described in table 2 to construct interaction terms with the different credit rating dummies following Kemper and Rao (2013):

$$NetDIss_{it} = \alpha_1 + \beta_0 CR_{POM} + \beta_1 EFN \cdot CR_{POM} + \beta_2 EFN + \phi K_{it} + u_{it} \quad (6)$$

$$NetDIss_{it} = \alpha_2 + \beta_3 CR_{Plus} + \beta_4 EFN \cdot CR_{Plus} + \beta_5 CR_{Minus} + \beta_6 EFN \cdot CR_{Minus} + \beta_7 EFN + \phi K_{it} + u_{it} \quad (7)$$

Table 8 displays the effect of probable credit rating changes on firms' net debt issuance when the external financing needs are also taken into account.

Table 8: Effect of probable change in credit rating on the net debt issuance vs. equity, sensitivity to external financing

Variables/statistics	Model 1	Model 2
Intercept	-0.1228***	-0.1228***
CR_{POM}	-0.0059* (0.0048)	
$EFN \cdot CR_{POM}$	-0.0007* (0.0005)	
CR_{Plus}		-0.0075* (0.0067)
$EFN \cdot CR_{Plus}$		-0.0007** (0.0004)
CR_{Minus}		-0.0043* (0.0035)
$EFN \cdot CR_{Minus}$		-0.0002 (0.0006)
EFN	0.0007* (0.0005)	0.0007* (0.0005)
F-statistics	6.2700	5.5500
Prob(F-statistics)	0.0000	0.0000
R-squared	0.0624	0.0624

Note: Standard errors are given within the parenthesis. ***, **, and * denote significance at the 1-, 5- and 10-% levels, respectively.

The results show that the coefficient for CR_{POM} is negatively significant as in the baseline model. Furthermore, we find that the interaction with EFN is negatively significant as well, which suggests that firms with greater external financing needs facing a probable up- or

downgrade are more likely to follow the CR-CS model when deciding on capital structure than other firms. In other words, these type of firms will issue less debt when close to a credit rating change compared to firms that are more likely to depend on internal funding. However, the effect on the net debt issuance from the interaction term is fairly small compared to what we had expected. A possible explanation to the small effect is that a credit rating can serve to reduce asymmetric information in the market as described in section 2.2, but firms that are highly reliant on external financing are quite transparent already such that they will be less concerned with information asymmetry. It could thus be argued that these type of firms have established reputations on the debt capital markets which allows them to raise their needed external funding, and a probable change in their credit rating will thus not affect this ability to raise funds (Kemper & Rao, 2013).

Now we turn to analyze the effects of real credit rating changes when the external financing needs are considered as well. The regression models to test this are extensions of equation (3), (4) and (5) from section 4.1.2 and table 9 below shows our results:

$$NetDIss_{it} = \alpha_3 + \beta_0 IGSG_{up} + \beta_1 EFN \cdot IGSG_{up} + \beta_2 IGSG_{down} + \beta_3 EFN \cdot IGSG_{down} + \beta_4 EFN + \phi K_{it} + u_{it} \quad (3)$$

$$NetDIss_{it} = \alpha_4 + \beta_5 BR_{up} + \beta_6 EFN \cdot BR_{up} + \beta_7 BR_{down} + \beta_8 EFN \cdot BR_{down} + \beta_9 EFN + \phi K_{it} + u_{it} \quad (4)$$

$$NetDIss_{it} = \alpha_5 + \beta_{10} NH_{up} + \beta_{11} EFN \cdot NH_{up} + \beta_{12} NH_{down} + \beta_{13} EFN \cdot NH_{down} + \beta_{14} EFN + \phi K_{it} + u_{it} \quad (5)$$

The results shows the same tendency as in the baseline models in table 5 i.e. that firms will issue more debt following an upgrade and issue less debt following a downgrade. However, what is interesting about these results is the fact that firms with larger external financing needs that have just been upgraded to investment grade (the interaction term $EFN \times IGSG_{up}$) will issue much *less* debt than firms relying more on internal funds. One potential explanation is that a firm with large external financing needs that have just received an investment grade would not risk being downgraded to speculative grade status again, since investment

grade comes with major benefits on the debt capital markets - particularly when a firm is relying significantly on external funding. They therefore chooses to decrease their debt with a significant amount. However, all other interaction terms are marginal and not very significant meaning that in general, firms with greater external financing needs do not react more strongly to real credit rating changes. Since the interaction terms for both probable and real rating changes overall are small and insignificant, H2 can in general be rejected thus it does not seem that firms with larger external financing needs will be more likely to have credit ratings in mind when deciding on capital structure.

Table 9: Effect of real credit rating changes on the net debt issuance vs. equity, sensitivity to external financing

Variables/statistics	Model 3	Model 4	Model 5
Intercept	-0.1252***	-0.1231**	-0.1216**
$IGSG_{up}$	0.0263*** (0.0107)		
$EFN \cdot IGSG_{up}$	-0.1531** (0.0892)		
$IGSG_{down}$	-0.0317*** (0.0105)		
$EFN \cdot IGSG_{down}$	-0.0035 (0.0171)		
BR_{up}		0.0131** (0.0063)	
$EFN \cdot BR_{up}$		0.0051 (0.0160)	
BR_{down}		-0.0265*** (0.0104)	
$EFN \cdot BR_{down}$		-0.0001* (0.0001)	
NH_{up}			0.0011 (0.0055)
$EFN \cdot NH_{up}$			0.0612* (0.0490)
NH_{down}			-0.0250*** (0.0086)
$EFN \cdot NH_{down}$			-0.0001 (0.0001)
EFN	0.0001 (0.0002)	0.0009** (0.0005)	0.0011** (0.0006)
F-statistics	7.6000	10.8300	11.5000
Prob(F-statistics)	0.0000	0.0000	0.0000
R-squared	0.0631	0.0632	0.0079

Note: Standard errors are given within the parenthesis. ***, **, and * denote significance at the 1-, 5- and 10-% levels, respectively.

4.2.3 Hypothesis 3: Commercial paper market

Now we want to test our third hypothesis, namely that firms participating in the commercial paper (CP) market are more sensitive to credit rating changes (H3).

To test the hypothesis we split the sample into two sub-samples: One sample with CP issuers, and a second sample consisting of non-CP issuers. We have used the variable "Standard & Poor's Short-Term Domestic Issuer Credit Rating"⁵ to indicate whether a firm is engaged in the CP market or not. The group of (non-)CP issuers are thus the set of firms that do (not) have a Short-Term Domestic Issuer Credit Rating on Compustat. Table 10 below shows our results of the impact on net debt issuance for probable credit rating changes for CP and non-CP issuing firms, respectively.

Table 10: Effect of probable credit rating changes on the net debt issuance, commercial (C) vs. non-commercial (NC) paper issuers.

Variables/statistics	C	C	NC	NC
Intercept	0.1974	-0.1195***	-0.1092***	-0.1097***
CR_{POM}	-0.0115* (0.0093)		-0.0005 (0.0032)	
CR_{Plus}		-0.0129* (0.0121)		0.0013 (0.0035)
CR_{Minus}		-0.0100* (0.0071)		-0.0022 (0.0039)
F-statistics	1.1300	1.0800	5.2500	4.6800
Prob(F-statistics)	0.3424	0.3705	0.0003	0.0003
R-squared	0.1190	0.1190	0.0046	0.0047

Note: Standard errors are given within the parenthesis. ***, **, and * denote significance at the 1-, 5- and 10-% levels, respectively.

It is clear how CP issuers react significantly to probable credit rating changes, whereas non-CP issuers do not. Furthermore, the CR-CS model seems to be a good indicator for the corporate financing behavior of commercial paper issuing firms, since these type of firms annually issue 1.2% less debt relative to equity when close to an up- or downgrade compared to firms not near a credit rating change. The effect on the net debt issuance is double the size of the effect in the baseline model in table 4, which supports the fact that a long-term credit rating change is much more important for firms on the CP market than firms not relying on this market as short-term capital. The findings also show that the significance of CR_{POM} is almost equally driven by the plus and minus segment i.e. the CP issuing firms will reduce their debt relative to equity both near an up- and downgrade, which further supports the CR-CS hypothesis. On the other hand, non-commercial paper issuing firms do not have probable credit rating changes in mind when deciding on capital structure, since the

⁵The variable is used for obligations with an original maturity of no more than 365 days-including commercial paper (S&P, 2022c).

coefficients are insignificant. The fact that firms participating in the CP market react more strongly to probable credit rating changes than non-CP issuing firms was expected, and H3 is thus confirmed. Now we want to test if this also applies to real credit rating changes, and the results can be seen in in table 11 below.

Table 11: Effect of real credit rating changes on the net debt issuance, commercial (C) vs. non-commercial (NC) paper issuers

Variables/statistics	C	C	C	NC	NC	NC
Intercept	0.1922	0.1930	0.2054	-0.1308***	-0.1289***	-0.1281***
$IGSG_{up}$	-0.0303 (0.0306)			0.0326*** (0.0097)		
$IGSG_{down}$	-0.1071*** (0.0448)			-0.0082 (0.0079)		
BR_{up}		0.0168* (0.0123)			0.0141** (0.0061)	
BR_{down}		-0.0581** (0.0288)			-0.0131* (0.0091)	
NH_{up}			-0.0066 (0.0106)			0.0105*** (0.0039)
NH_{down}			-0.0544** (0.0280)			-0.0134** (0.0058)
F-statistics	3.9900	3.6500	5.7600	8.9300	8.2300	10.0100
Prob(F-statistics)	0.0014	0.0028	0.000	0.0000	0.0000	0.0000
R-squared	0.1213	0.1209	0.1207	0.0067	0.0069	0.0074

Note: Standard errors are given within the parenthesis. ***, **, and * denote significance at the 1-, 5- and 10-% levels, respectively.

These findings show that CP issuing firms do not alter their capital structure following an upgrade, but they react by reducing their net debt strongly following a downgrade. This is indicated since the coefficients of "up" are insignificant, whereas the coefficients of "down" are strongly significant. A firm downgraded to high yield will for example issue 10.7% less net debt annually relative to equity, but an upgrade to investment grade will not affect the capital structure of a CP issuing firm. This is aligned with minimum credit rating targeting (as described in section 4.1.1). Even after a notch rating downgrade (NH_{down}) a firm will reduce its net debt by 5.4%, which is a much larger effect than in the baseline model in section 4.1.2. This can be explained by the fact that even a notch credit rating downgrade can have large discrete costs for a CP issuing firm, since this change in the long-term rating can affect the short-term rating and thus the access to the CP market. A firm will thus try to move into a higher credit rating - or avoid another downgrade - by reducing its net debt. In general, CP issuing firms react much more strongly to real downgrades than non-CP issuing firms, but at the same time firms on the CP market seem more reluctant to issue debt

after an upgrade compared to non-CP issuing firms that increases their net debt issuance by a significant amount. One could argue that this is because the CP issuing firms are more concerned than the non-CP issuing firms about being downgraded as a result of new debt issues (cf. CR-CS theory), and they therefore chooses not to increase their debt relative to equity after an upgrade. H3 can thus only be partly confirmed, since CP issuing firms react more (less) strongly to downgrades (upgrades) than non-CP issuing firms.

4.2.4 Hypothesis 4: Growth opportunities

The fourth hypothesis is that firms with growth opportunities may be more concerned about credit rating changes than firms without the same opportunities, since these type of companies need funding in order to grow (H4). Hence, they are more likely to be active debt capital market participants.

To test the hypothesis we use the approximation of Tobin's Q by Chung and Pruitt (1994) to assess if a firm has growth opportunities. We split the sample into two sub-samples: One sample with "high" Q (above 1) firms and a second sample with "low" Q (below 1) firms. Table 12 below shows the results of the impact of probable credit rating changes on the net debt issuance when growth opportunities are taken into account.

Table 12: Effect of probable credit rating changes on the net debt issuance, growth opportunities.

Variables/statistics	High Q	High Q	Low Q	Low Q
Intercept	-0.1602***	-0.1601***	-0.0435***	-0.0437***
CR_{POM}	-0.0065* (0.0046)		0.0015 (0.0020)	
CR_{Plus}		-0.0056* (0.0048)		0.0030* (0.0023)
CR_{Minus}		-0.0075* (0.0062)		0.0000 (0.0023)
F-statistics	8.0600	6.6100	2.8200	2.6000
Prob(F-statistics)	0.0000	0.0000	0.0237	0.0239
R-squared	0.0046	0.0046	0.0013	0.0013

Note: Standard errors are given within the parenthesis. ***, **, and * denote significance at the 1-, 5- and 10-% levels, respectively.

The findings indicate that the CR-CS hypothesis holds for firms with large growth opportunities such that these type of firms will issue 0.65% less debt relative to equity when near a credit rating change compared to firms not facing a probable credit rating change. As in

H3, the result is driven by both the minus and plus segment, which is further supportive of the CR-CS hypothesis. The results also show that firms with low growth opportunities do not take probable credit rating changes into account when deciding on capital structure, since the coefficients are insignificant. It can thus be concluded that H4 holds since growth opportunities influence a firm's decision to follow a capital structure policy with probable credit ratings in mind. These results were expected from theory, and we will now examine if our hypothesis still holds when looking at real credit rating changes.

Table 13: Effect of real credit rating changes on the net debt issuance, growth opportunities

Variables/statistics	High Q	High Q	High Q	Low Q	Low Q	Low Q
Intercept	-0.0447***	-0.0426***	-0.0406**	-0.2025***	-0.2030***	-0.2040***
$IGSG_{up}$	0.0150 (0.0120)			0.0280** (0.0119)		
$IGSG_{down}$	-0.0029*** (0.0063)			-0.0393 (0.0063)		
BR_{up}		0.0075* (0.0050)			0.0141* (0.0090)	
BR_{down}		-0.0207*** (0.0039)			-0.0140 (0.0190)	
NH_{up}			0.0063** (0.0028)			0.0098** (0.0029)
NH_{down}			-0.0189*** (0.026)			-0.0112 (0.0025)
F-statistics	11.9200	11.6500	12.5900	7.5500	8.9900	15.4300
Prob(F-statistics)	0.0000	0.0000	0.000	0.0000	0.0000	0.0000
R-squared	0.0038	0.0039	0.0037	0.0028	0.0039	0.0057

Note: Standard errors are given within the parenthesis. ***, **, and * denote significance at the 1-, 5- and 10-% levels, respectively.

Our research in table 13 shows that firms with large growth opportunities adjust their capital structure more following a real downgrade compared to firms with less growth opportunities, but at the same time they adjust their capital structure less following a real upgrade. These are the same results as found in H3 for real rating changes, and the explanation is thus the same: Firms with large growth opportunities will after a downgrade strive to achieve an upgrade or avoid another downgrade by issuing less debt. At the same time, they are more reluctant to issue new debt after an upgrade, since they will not risk a downgrade. However, the difference in the adjustments of capital structure for high growth versus low growth companies are relatively small - much smaller than the difference between CP and non-CP issuing firms. The results are thus not as significant and should be interpreted with caution. We can thereby not fully confirm H4 concerning firms with growth opportunities.

The results from our four hypotheses thus show that the relationship between credit ratings

and capital structure is conditional on firms' financial circumstances. Particularly, we find that there is a difference between IG and SG firms (H1) and CP and non-CP issuing firms (H3). The difference between firms with more or less external financing needs and growth opportunities is less significant (H2 & H4)

5 Discussion and interpretation

5.1 Empirical results and shortcomings

The results of our research are prone to some uncertainty as our analysis has a few shortcomings, which we will touch upon in this section.

First of all, one could argue that plus and minus ratings are not an exact measure for a probable upgrade or downgrade, respectively. We considered S&P's Creditwatch or Rating Outlook as well for determining whether firms are near a rating change. CreditWatch highlights the rating agency's opinion regarding the potential direction of a short-term or long-term rating, and it focuses on specific events (such as a merger, recapitalization, or regulatory action) and short-term trends that cause ratings to be placed on CreditWatch. A S&P Ratings Outlook assesses the potential direction of a long-term credit rating over the intermediate term (six months to two years), and it focuses on any changes in the economic and/or fundamental business conditions (S&P, 2018). However, both a CreditWatch listing and a Rating Outlook do not mean that a rating change is inevitable, and rating changes may also occur without the ratings having first appeared on CreditWatch/Outlook. Because of this, we did not find these type of distinctions to be a better measure for a probable rating change than the plus and minus ratings. Overall, one can thus conclude that it is hard to find an exact measure for probable rating changes, and this can also help to explain why the effect from a probable rating change on the net debt issuance is much smaller than the effect from a real rating change on the capital structure decisions. Our results are aligned with Krichene and Khoufi (2015) who also find that there is a larger effect from real rating changes than from a probable rating changes.

Furthermore, a possible complicating factor in our paper is that we use the credit ratings at the beginning of a year. If the rating situation for a firm changes during the year, our measure

can become inaccurate. However, this is not a problem that can be avoided, since we only have access to credit ratings and other fundamental variables such as debt and equity on an annual basis. The analysis would improve if we had data on a higher frequency, but to our knowledge this is not possible. Lastly, adjustments to capital structure take time to execute such that there is possibly a significant lag between the time a capital structure decision is made and the time it appears in the books. These factors add noise to the empirical tests resulting in the fact that the estimated coefficients may underestimate the true credit rating effects on capital structure - and this applies to probable as well as real credit ratings. Kisgen (2006) mentions this as well, so it seems to be a common shortcoming in the literature.

It is also commonly known that credit ratings before the financial crisis in 2007-08 failed to accurately portray default risk, and studies such as DeHaan (2017) find that market participants have decreased their reliance on credit ratings after the crisis. An explanation to this is that rating agencies suffer spill over reputation damage from their biased ratings during the crisis. This could potentially have an effect on our results, if firms know that market participants such as investors rely less on credit ratings. For future research, one could split the sample in smaller periods to examine if there are differences in the effect of credit rating changes on capital structure due to differences in the reliance on credit ratings.

5.2 Applying our results to the real world

It has been mentioned intuitively by practitioners for years that maintaining or achieving a specific credit rating is appealing for a firms' capital structure decisions. However, this link between credit ratings and capital structure has only been confirmed empirically in later years starting with Kisgen's (2006) CR-CS theory. Our research first of all extends this analysis to more recent years, and we further highlight the difference in firm behaviour between facing a probable and real credit rating change. The results therefore emphasize the importance of credit ratings for capital structure behavior, which can be interesting for several financial actors. The firms - and especially the CFOs - are required to make very essential decisions for firm value that can be impacted by credit ratings. We further find that firms should not only focus on being rated or not, but the firms should also consider the within-rating scales. Our results indicate that different ratings and rating changes (to investment or speculative

grade, broad, or notch) can potentially affect crucial factors to the overall firm value such as cost of debt, the accessibility to financial markets, and information asymmetry. At the same time, capital structure policy must be designed to fit the financial circumstances of a particular firm. For example, our results show that CP issuing firms react much stronger to both probable and real credit rating changes compared to non-CP issuing firms, and this highlights that firms which are not on the CP market may well attach a much smaller value to credit ratings. On the other hand, firms that rely heavily on the CP market will almost certainly work hard to maintain high long-term credit ratings.

Furthermore, consultants and investment banks advising on capital structure will also benefit from our paper by being up to date on the newest research on firm's capital structure behaviour in order to give the best advise. It can also be useful for rating agencies to be aware of how their setting of credit ratings impact managers' corporate finance decision making, and it might help the rating agencies improve their framework.

Lastly, the current global economic situation makes credit ratings even more important. The world is still fighting the aftermath of Covid-19, and BlackRock (2020) describes how ratings downgrade activity in investment grade corporate bonds and fallen angel volumes have increased significantly. This spike is similar in magnitude to other periods with stressed market conditions such as the Global Financial Crisis in 2008, the US Credit Rating Downgrade in 2011, and the global oil sell-off in 2016. Furthermore, Western central banks are pushing interest rates to levels not seen in 15 years and shrinking their balance sheets. Market actors are selling off their corporate bonds in favor of safer assets such as Treasury bonds. This has resulted in quickly decreasing prices of corporate bonds, especially for less creditworthy firms: Yields on speculate grade bonds have soared to 9.4% from 4.4% in January in the US. Furthermore, the yield spread between speculative grade bonds and bonds with the lowest investment grade (BBB-) has doubled the past year due to inflation and the risk of recession (Economist, 2022). This shows how much more expensive it is to borrow for speculative grade firms compared to investment grade firms, which indicate that credit ratings should play a major factor in firms' capital structure decisions at the moment.

For future research, it would thus be interesting to conduct the same analysis as in this

paper on the time period post Covid-19 characterized by a high yield-high inflation regime. One would expect that firms would react even stronger to credit rating changes than our results show by taking more defensive steps to limit the downgrade consequences due to the increasing yields.

6 Conclusion

The purpose of this paper is to provide clarity regarding the relationship between credit ratings and capital structure, and in particular testing the CR-CS hypothesis by Kisgen (2006). Using a fixed effects OLS approach, we find that both probable and real credit rating changes have a significant effect on capital structure decisions. In our baseline scenario, the CR-CS hypothesis partly holds for probable rating changes, since firms will lower their net debt issuance when close to a downgrade, whereas a probable rating upgrade does not seem to affect corporate finance decisions. This suggests that firms target minimum credit ratings over time. Furthermore, real credit rating changes show asymmetric effects such that firms issue less debt following a downgrade, and they issue more debt following an upgrade which can be explained by the costs/benefits and regulation requirements associated with rating changes. Our findings further indicate that the effect of a real rating change on net debt issuance is largest when the firm is either upgraded to investment grade or downgraded to speculative grade compared to a broad or notch credit rating. Overall, we show that firms react differently to probable and real credit rating changes.

In the second part of our analysis we test four hypotheses concerning what attributes make some firms more or less concerned about their credit rating when deciding on capital structure. We can conclude that particularly investment grade firms and commercial paper issuing firms react much stronger to both probable and real rating changes than speculative grade firms and non-commercial paper issuing firms, respectively. This was expected from theory, since investment grade firms and firms on the commercial paper market benefits a lot from maintaining a high credit rating. Besides these two firm attributes, we also explore if external financing needs and growth opportunities have a direct impact on credit ratings' importance for firms' capital structure, but these results were not as significant.

Overall, our findings indicate that credit ratings are a factor which firms should consider when deciding on capital structure, especially due to the current world situation, and especially for firms with an investment grade and/or firms dealing on the commercial paper market. We hope that this paper will give inspiration to future studies, since the relationship between capital structure and credit ratings is still a relatively uncovered topic.

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7 Appendix

Definition of variables used for construction of the dependent and independent variables in the regressions

D_{it} = Book long-term debt plus book short-term debt for firm i at time t .

ΔD_{it} = Long-term debt issuance minus long-term reduction plus changes in current debt for firm i from time $t-1$ to t .

E_{it} = Book value of shareholders' equity issuance for firm i at time t .

ΔE_{it} = Sale of common and preferred stock minus purchases of common and preferred stock for firm i from time $t-1$ to t .

A_{it} = Beginning-of-year total assets for firm i at time t .

$$EFN = \text{Real growth rate of assets} - \text{Sustainable growth rate} = \frac{A_t - A_{t-1}}{A_{t-1}} - \text{ROE} \times \text{Retention rate} = \frac{A_t - A_{t-1}}{A_{t-1}} - \frac{\text{Net income at time } t}{\text{Avg. shareholders' equity at time } t} \times \frac{EPS - DPS}{EPS}$$

The sustainable growth rate is the rate at which the firm can grow while using its internal revenue without relying on external financing needs. It can be computed by multiplying the firm's return of equity (ROE) with the retention rate, which is the proportion of earnings kept back in a firm to grow the business as opposed to being paid out as dividends to shareholders. EPS and DPS is earnings per share and dividends per share, respectively. "